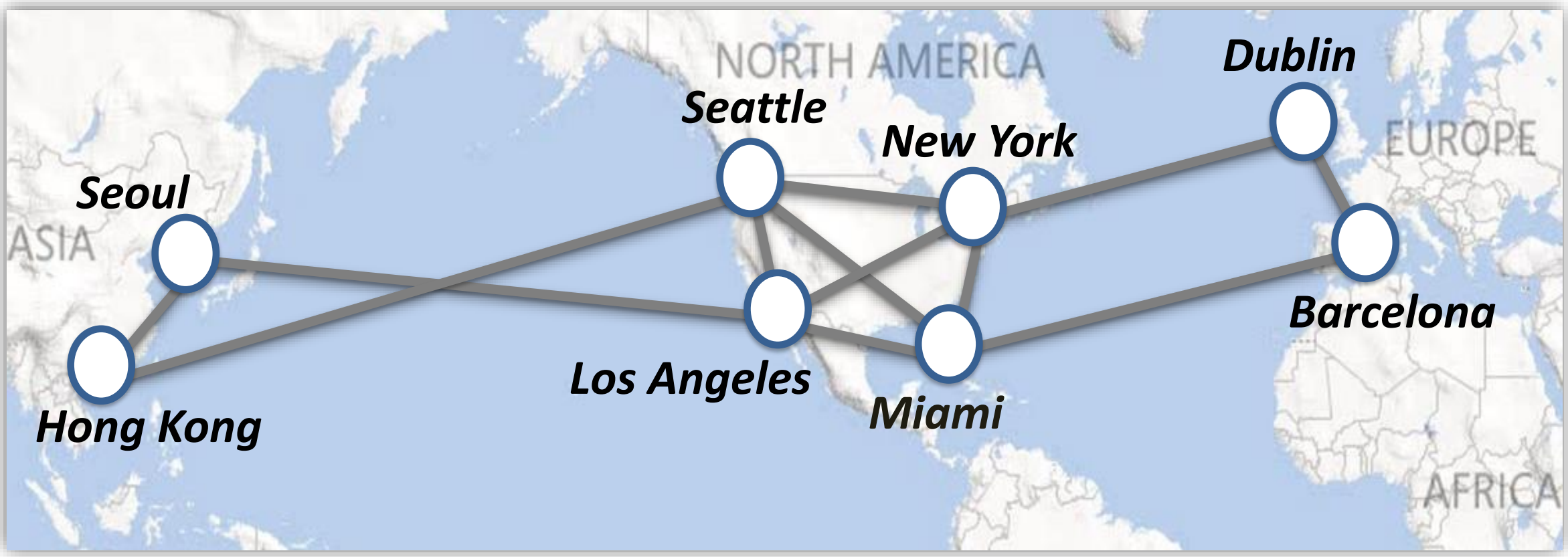
SWAN: Software-driven wide area network

Ratul Mahajan

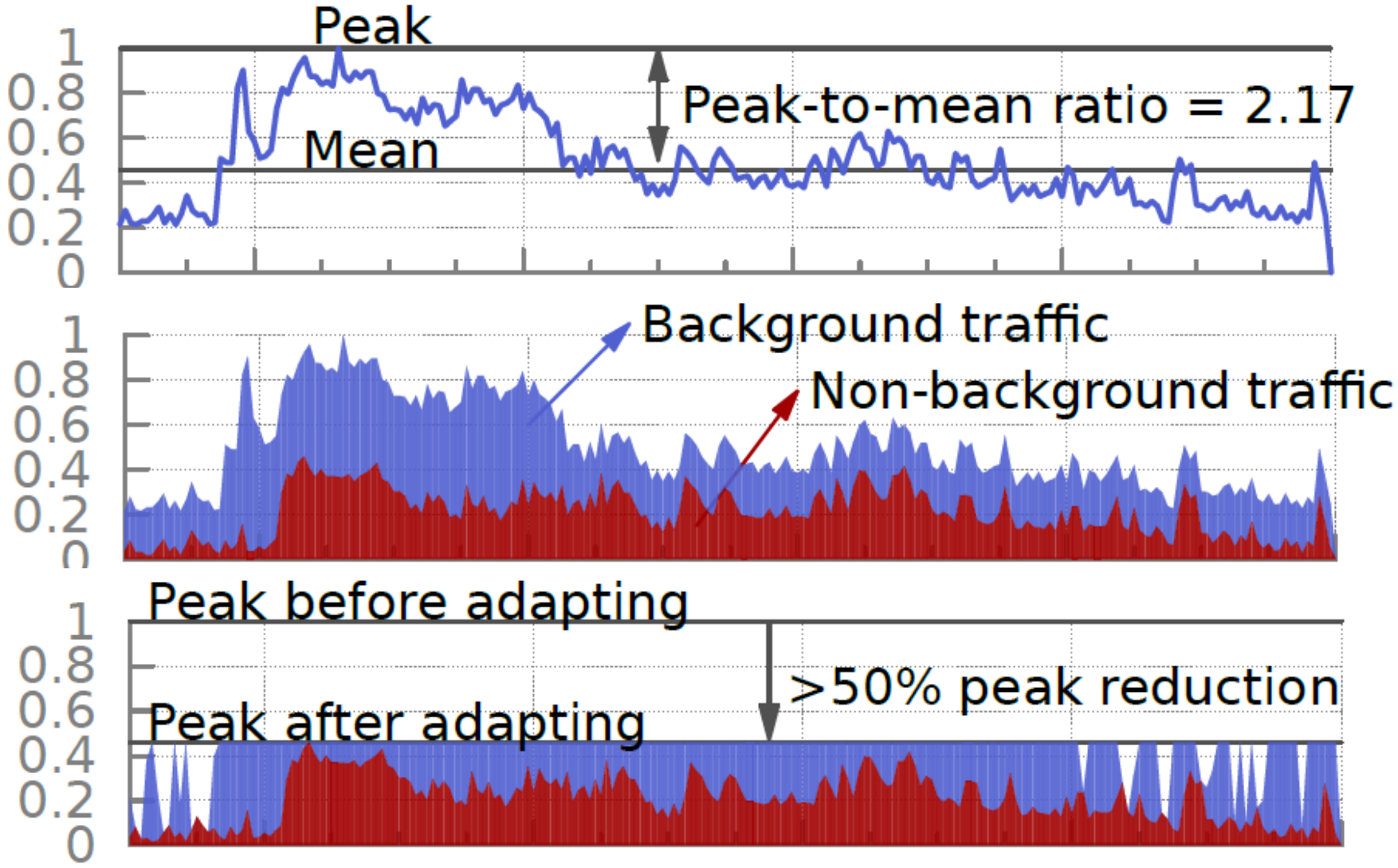
Microsoft Research

Inter-DC WAN: A critical, expensive resource

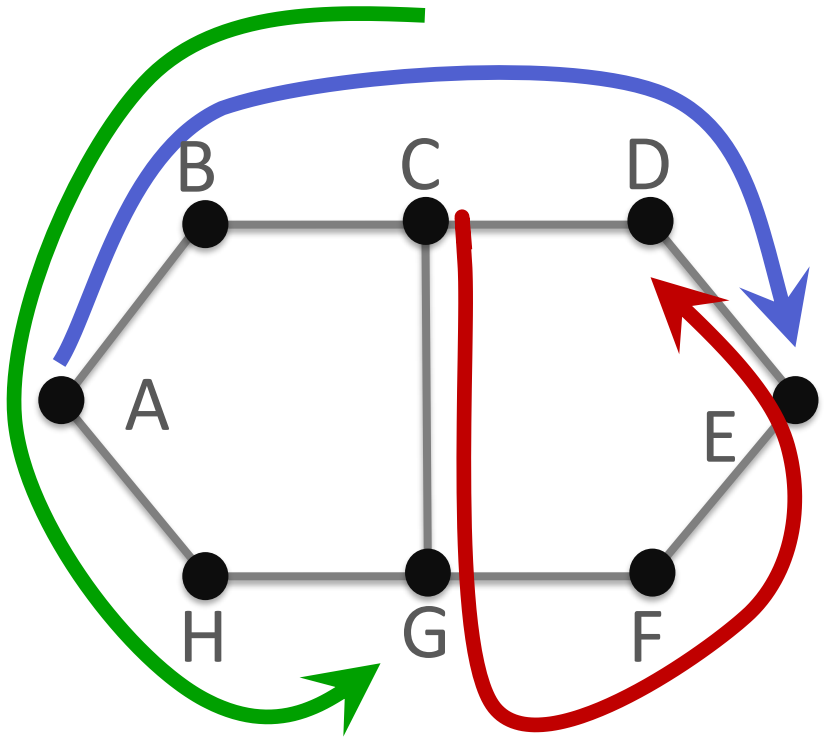


But it is highly inefficient

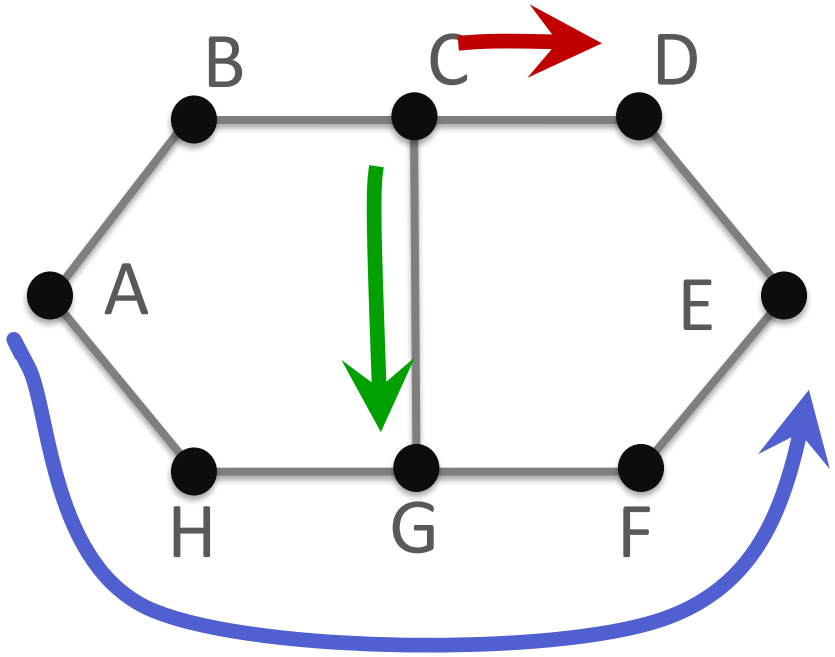
One cause of inefficiency: Lack of coordination



Another cause of inefficiency: Local, greedy resource allocation



Local, greedy allocation



Globally optimal allocation

SWAN: Software-driven WAN

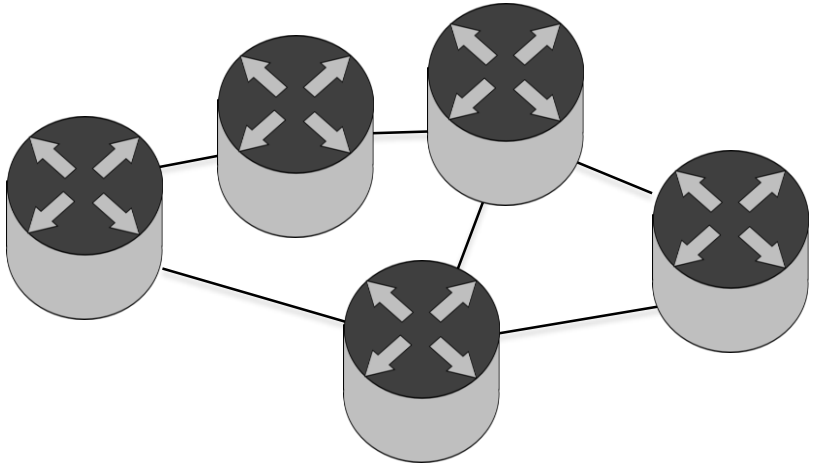
Goals:

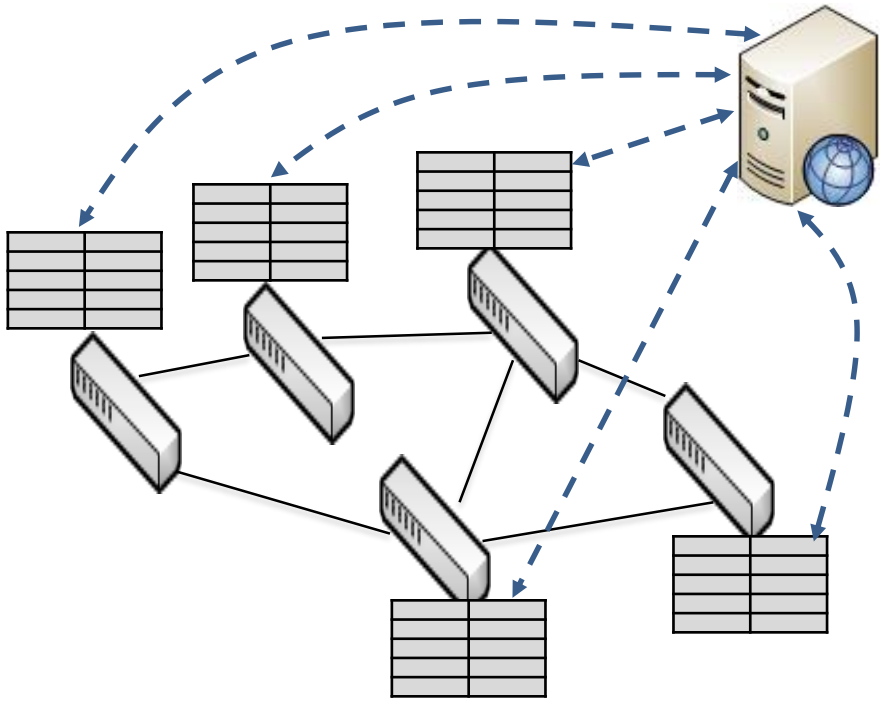
- Highly efficient WAN
- Support flexible sharing policies
  - Strict priority classes
  - Max-min fairness within a class

Key design elements:

- Coordinate the sending rate of services
- Centralized resource allocation

SDN primer





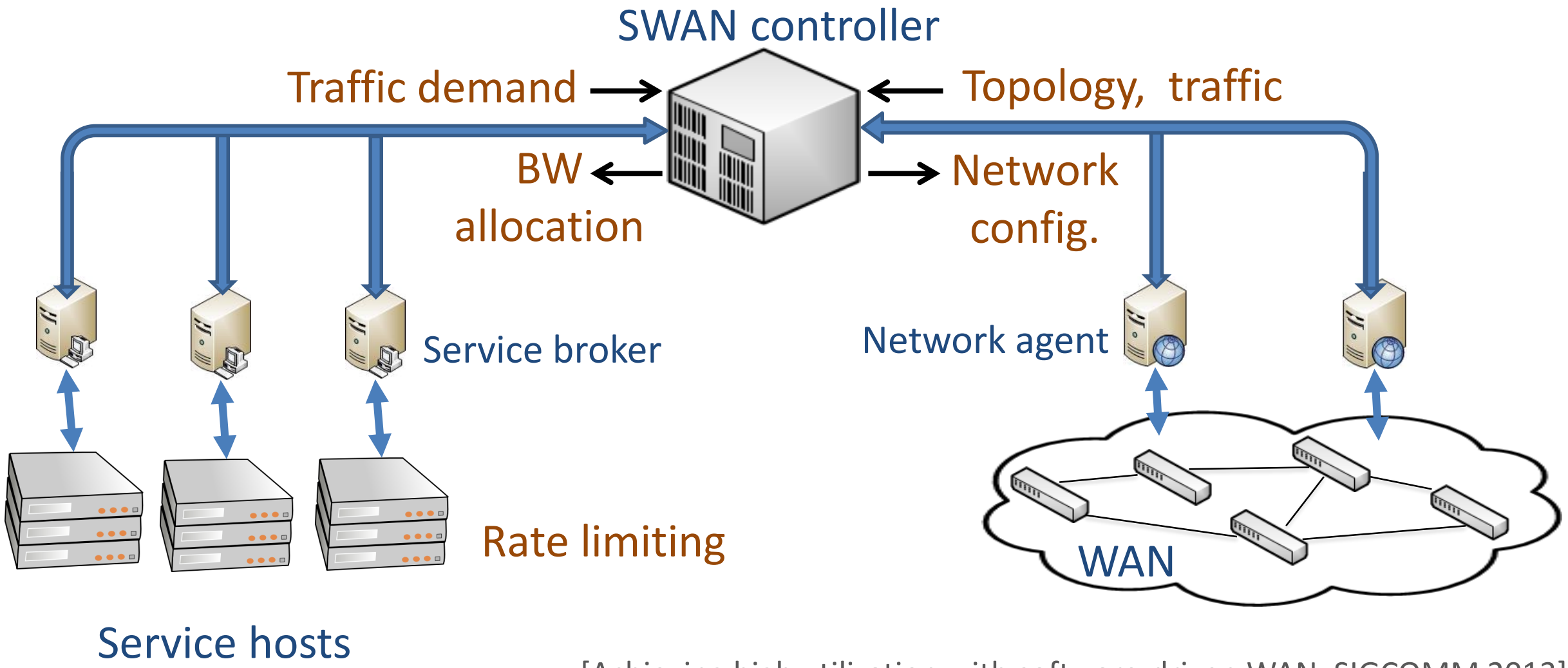
Networks today

- Beefy routers
- Control plane: distributed, on-board
- Data plane: indirect configuration

SDNs

- Streamlined switches
- Control plane: centralized, off-board
- Data plane: direct configuration

SWAN overview



[Achieving high utilization with software-driven WAN, SIGCOMM 2013]

Key design challenges

Scalably computing BW allocations and network config

Avoiding congestion during network updates

Working with limited switch memory

Scalably computing allocation

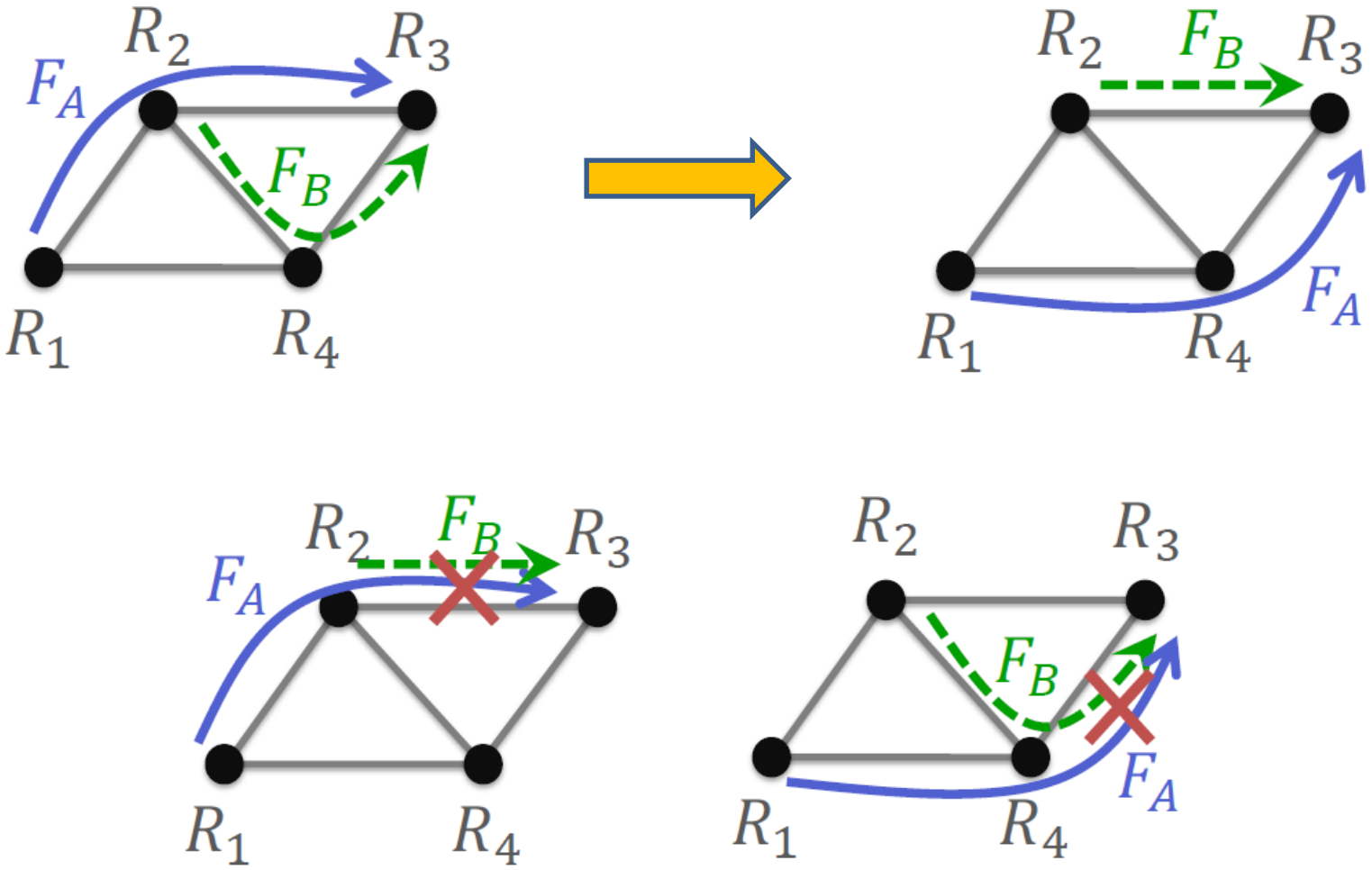
Path-constrained, multi-commodity flow problem

- Allocate higher-priority traffic first
- Fair within a class (weighted, max-min)

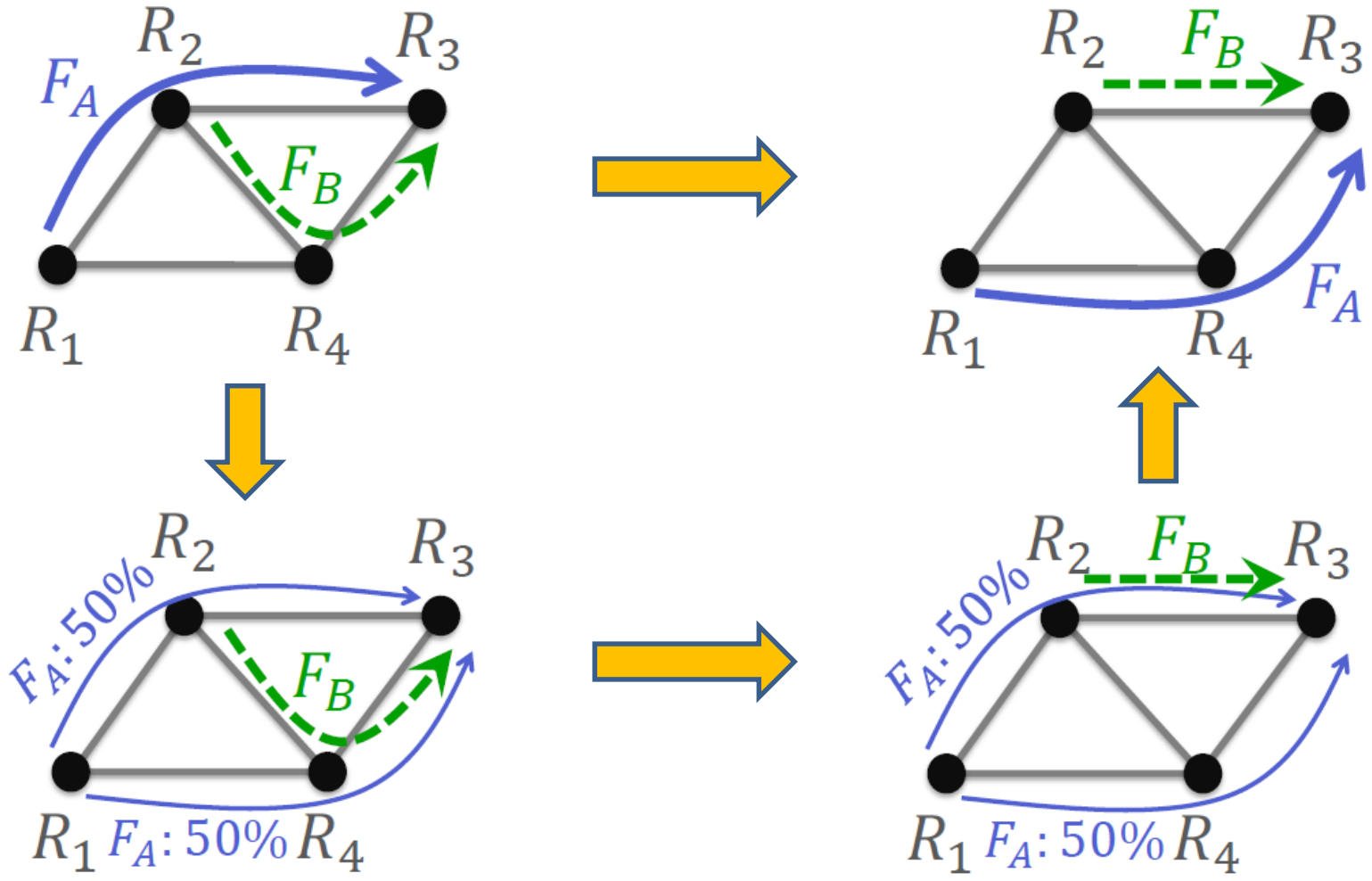
Solve at the granularity of DCs

- Split DC-level allocation fairly among services
- Derive switch configuration by leveraging network symmetry

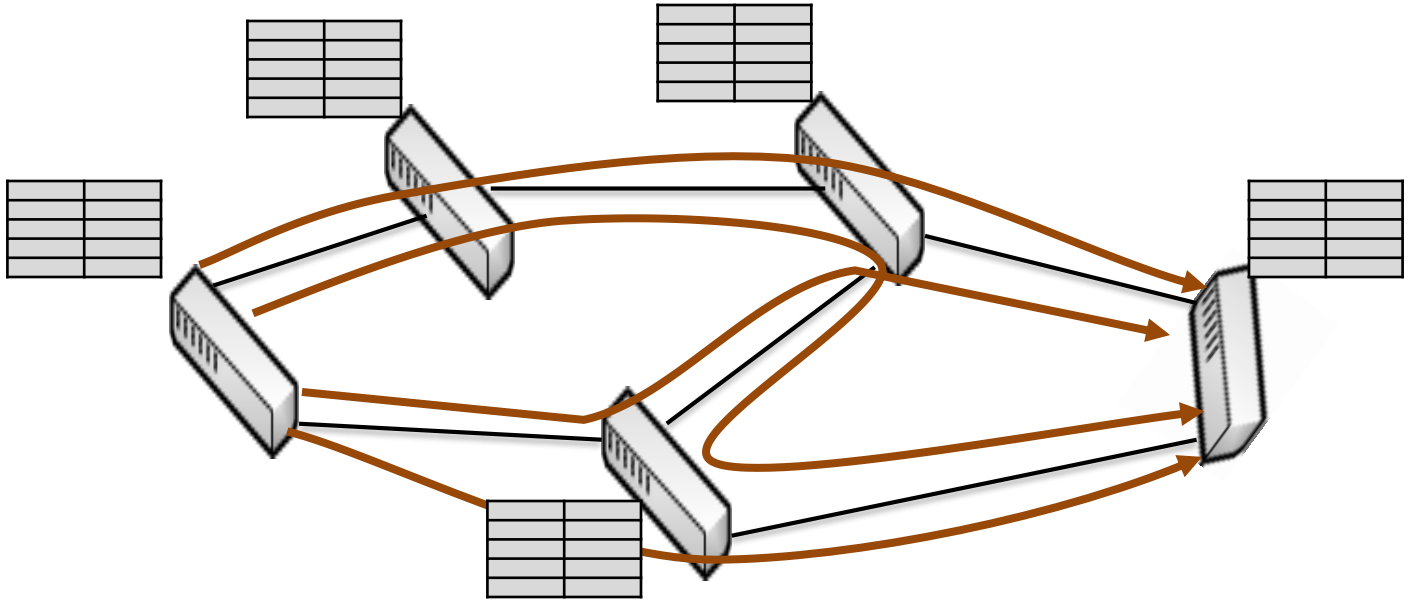
Congestion during network updates



Congestion-free network updates



Working with limited switch memory

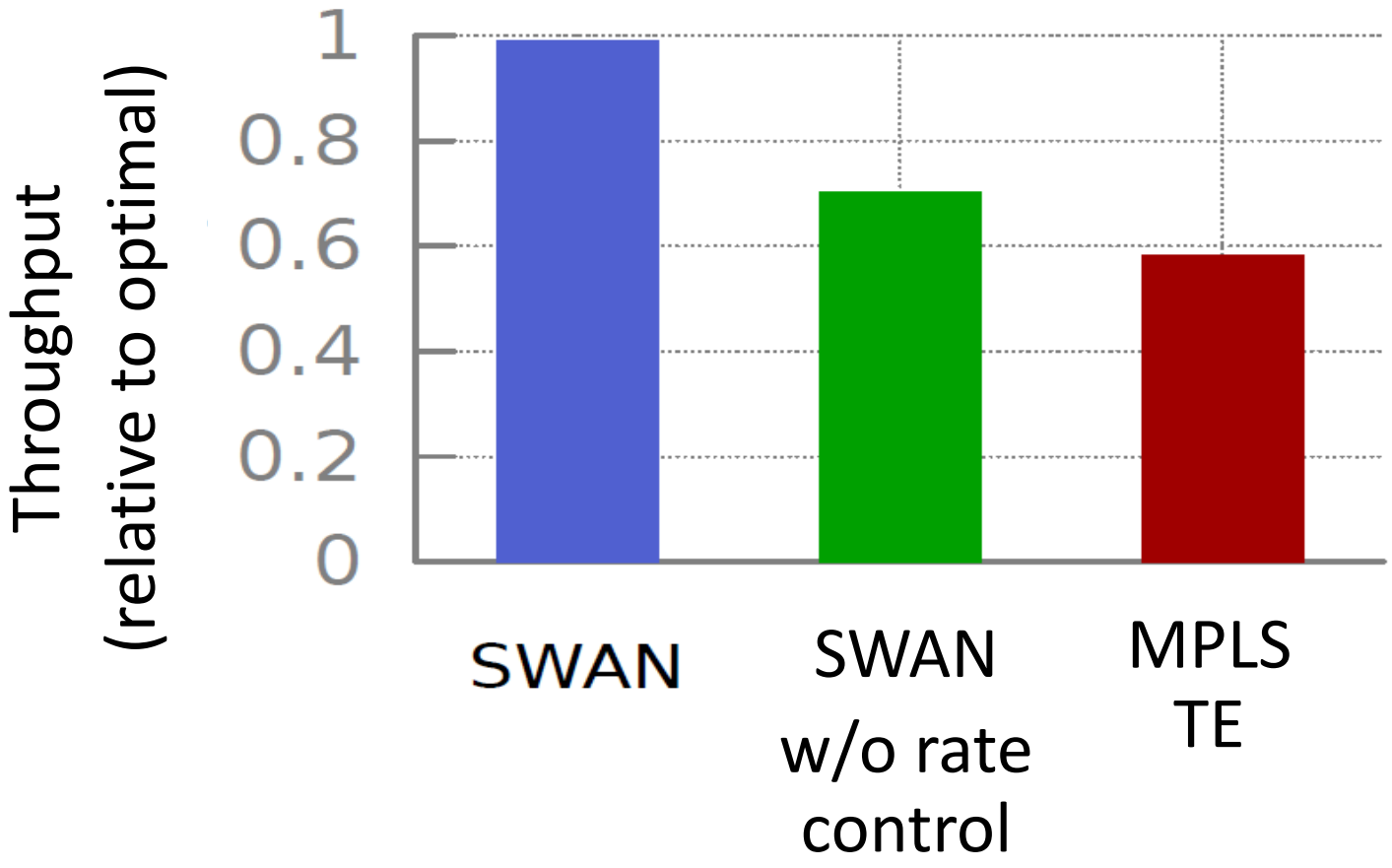


Use tunnel-based forwarding

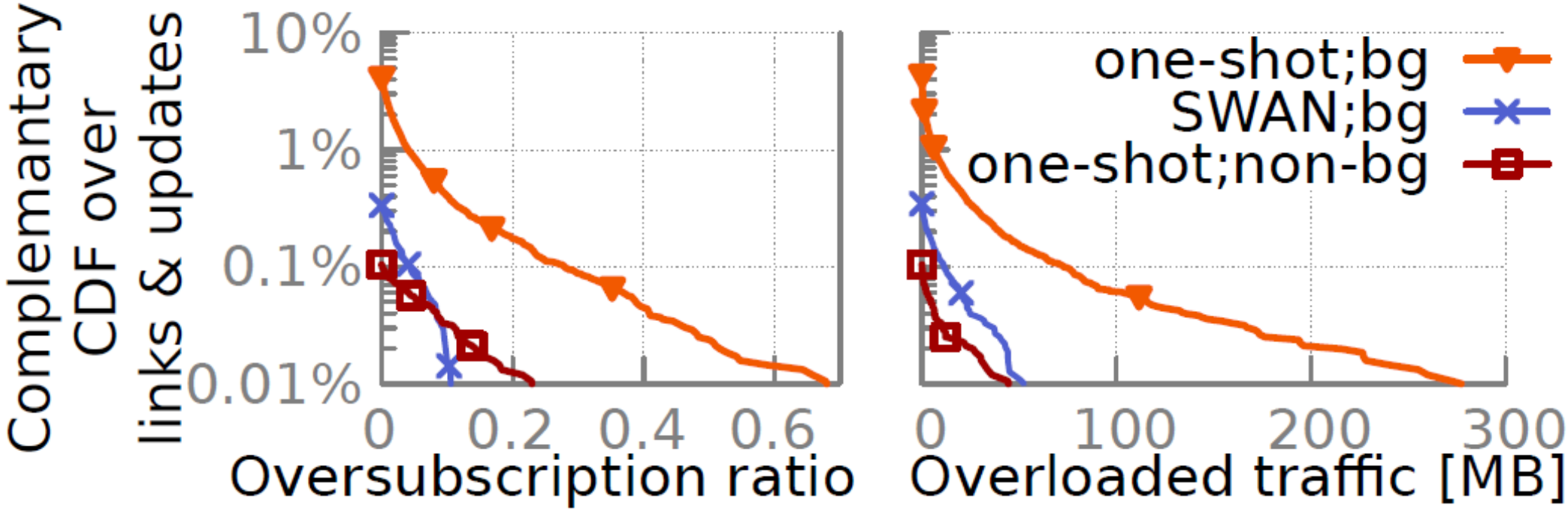
Install only the “working set” of tunnels

– Efficient mechanisms to update the set

SWAN comes close to optimal



Network updates: SWAN provides congestion-controlled updates



Summary

SWAN yields a highly efficient and flexible WAN

- Coordinates transmissions of services
- Allocates resources centrally
- Manages transitions by using scratch link and memory capacity

High efficiency is key to cost-effective cloud services

- Many avenues for impactful research
- Opportunity to be “clean slate”